THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT
ANNUAL JOINT EXAMINATION

FORM THREE
032 CHEMISTRY

Time: 3 Hours **November 2024**

INSTRUCTIONS

1. This paper consists of sections A, B, and C with total **eleven (11)** questions.
2. Answer all questions in section A and B and **two (2)** questions from section C.
3. Section A carry **Sixteen(16)** marks, Section B carry **fifty four (54)** marks and Section C carry **thirty (30)** marks
4. Cellular phones and any unauthorized materials are not allowed in examination room.
5. Write your **examination number** on every page of your answer sheet(s)
6. The following constants may be used:-
   Atomic mass: H=1, C=12, N=14, O=16, Na=23, Ca = 40, S=32, Mg=24, Cl=35.5, K =, Pb = 207, Fe = 56
   Avogadro's number = $6.02 \times 10^{23,}$ GMV at S.T.P = $22.4dm^3$, 1Faraday = 96,500coulombs.
   Standard pressure = 760mm Hg, Standard temperature = 273K, 1 litre = $1dm^3 = 1000cm^3$.

SECTION A (16 marks)

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter besides the item number in the answer sheet provided

i. Which of the following is not the purpose of administering firs aid?
   A. To relieve pain
   B. To bring hope to the victim.
   C. To prevent permanent disability
   D. To bring a dead person back to life
   E. To reduce the possibility of death.

ii. The gas law which relates the volume of a gas to moles of the gas is:
   (A) Avogadro's Law
   (B) Gay-Lussac's Law
   (C) Boyle's Law
   (D) Charle's Law

iii. In the following ionic equations:
   $Ca^{2+}_{(aq)} + 2Cl^-_{(aq)} + 2Ag^+_{(aq)} + 2NO_3^-_{(aq)} \rightarrow Ca^{2+}_{(aq)} + 2NO_3^-_{(aq)} + 2AgCl_{(s)}$
   Which of the following chemical species are called spectator ions?
   A. $Ca^{2+}_{(aq)}$ and $Ag^+_{(aq)}$

B. $Ca^{2+}_{(aq)}$ and $Cl^{-}_{(aq)}$

C. $Cl^{-}_{(aq)}$ and $Ag^{+}_{(aq)}$

D. $Ca^{2+}_{(aq)}$ and $NO_3^{-}_{(aq)}$

E. $NO_3^{-}_{(aq)}$ and $AgCl_{(s)}$

iv. Which of the following is not true about oxygen?

A. Oxygen supports combustion.
B. It is a very strong oxidizing agent.
C. It burns like a fuel
D. It reacts with non-metals to form acidic oxides.
E. It reacts vigorously with many metals to form basic oxides

v. If a sample of matter is uniform throughout and cannot be separated into different substances by physical means, it is:

A. A compound
B. Either a compound or an element
C. A homogeneous mixture
D. A heterogeneous mixture
E. An element

vi. During the electrolysis of acidified water which of the following takes place:

(A) Oxygen is released at cathode
(B) Oxygen is released at anode
(C) Hydrogen is released at anode
(D) Sulphur dioxide is released at anode

vii. The sign on a sulphuric acid bottle says it contained 49 grams of sulphuric acid; that equals to how many moles of the sulphuric acid?

A. 0.5
B. 49
C. 1
D. 0.75
E. 2

viii. Substance X liberates chlorine gas from acidified potassium chloride. What is the behavior described by X

A. An oxidizing agent
B. Catalyst
C. An oxidizing and reducing agent
D. A reducing agent
E. Bleaching agent

ix. Elements belonging to the same group have similar properties because:

A. They have similar electronic configuration in valence shell
B. Their atomic numbers increases as we move down a group
C. They are all metallic or non-metallic elements
D. Their number of electrons increases steadily

x. The process of giving away water of crystallization to the atmosphere by a chemical substance is called
A. Vaporization
B. Deliquescence
C. Hygroscopic
D. Sublimation
E. Efflorescence

2. Match the items in LIST A with the corresponding items in LIST B by writing the letter of correct response besides the item number in the answer sheet provided

List A	**List B**
(i) A homogenous mixture whose concentration is known	A. Titrant
(ii) The estimation of the concentration of a solution of an acid by reacting the acid with a standard alkali solution.	B. Titrate
(iii) A solution added to another solution whose concentration has to be determined	C. Standard solution
(iv) Substance to be analyzed or determined	D. Concentration
(v) The point at which indicator changes its colour	E. Molarity
(vi) A defined quantity of substance in a define volume of solution	F. End point
	G. Titration
	H. Indicator
	I. Alloy

SECTION B (54 marks)

3. (a) What is the purpose of using warning signs on the containers of chemicals in the laboratory?
(b) Compare the properties of covalent and electrovalent compounds on the following points:
(i) State at room temperature
(ii) Melting and boiling points
(iii) Solubility
(iv) Electrical conductivity

4. (a) i. Distinguish between electrochemical equivalent and chemical equivalent.
ii. State Faraday's Second Law of electrolysis
(b) A current of 5 amperes was passed through two electrolytes connected in series, one containing silver nitrate and the other containing copper sulphate solutions, and caused deposition of 6.4g of copper at the cathode. Calculate the mass of silver deposited at the cathode in the cell with silver nitrate electrolyte.

5. (a) Briefly explain two amendments made on Daltons atomic theory
(b) Boron has two naturally occurring isotopes: ^{10}B with an atomic mass of 10.01 amu and the percentage abundance of 20%, and ^{11}B with an atomic mass of 11.01 amu and the percentage abundance of 80. If you were asked to construct a periodic table, what atomic mass would you show for Boron on that periodic table?

6. The equation for the action of heat on calcium nitrate is:

$2(NO_3)_{2(s)} \longrightarrow 2CaO_{(s)} + 4NO_{2(g)} + O_{2(g)}$

(i) How many moles of NO_2 are produced when 1 mole of $(NO_3)_2$ decomposes?
(ii) What volume of O_2 at S.T.P will be produced on heating 65.6 g of $(NO_3)_2$?
(iii) Find out the mass of CaO formed when 65.6 g of $(NO_3)_2$ is heated.

(iv) Find out the mass of $(NO_3)_2$, required to produce 5 moles of gaseous products.
(v) Find out the mass of $(NO_3)_2$ required to produce 44.8litres of NO_2 at S.T.P.

7. A solution of sodium hydroxide is added to a solution of hydrochloric acid.
   (a) Given that both solutions are at room temperature what type of reaction takes place if the final temperature of the reaction mixture is 36°C?
   (b) Draw an energy level diagram for the reaction
   (c) State four application of the reaction above in our daily life

8. (a)Distinguish between titrant from analyte as applied in volumetric analysis.
   (b) The hydroxide of a metal M reacts with hydrochloric acid as follows:

$$M(OH)_2\ (_{aq}) + 2HCl\ (_{aq}) \rightarrow MCl_2(_{aq}) + 2H_2O\ (_l)$$

   It was found out that 5.8 g of M(OH)2 reacted with 0.2 M hydrochloric acid. Calculate the
   (i) Molar mass of $M(OH)_2$
   (ii) Atomic mass of M
   (iii) Suggest the name of the metal M.

SECTION C (30 marks)

9. (a) Iron and aluminium both are found in similar oxides; Fe_2O_3 and Al_2O_3 but different methods for extraction are used. Explain why the methods for extraction are different.
   (b) By using diagram describe the extraction of iron from the haematite ore by writing only the chemical equations for the reaction involved in the extraction.

10. A doctor at Seypalm hospital prescribed antacid tablets made from magnesium hydroxide to a student who was suffering from excess stomach acid.
    (a) Why did the doctor prescribe the antacid tablets?
    (b) Write a balanced chemical equation that took place in the student's stomach after taking the tablets.
    (c) If the normal acid content in the stomach is 160.00 millimolar and the stomach secretes 0.50 litres of gastric juice in a day. Calculate the mass of the tablets to be given to the student if the gastric juice had 210.00 millimolar hydrochloric acid
    (d) If each tablet of antacid weights 145mg of magnesium hydroxide, how many tablets will the student be given?
    (e) If a dose requires 1 tablet per every 12 hours, how many days should the student take to finish the dose?

11. Fuels differ greatly in quality. There are certain characteristics that make a good fuel. In eight points, discus those characteristics.